Gender figures in math: Do female math teachers have an impact on female students' outcomes?

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Abstract

Women's underrepresentation in STEM is partly rooted in math performance gaps that emerge early and persist. This paper studies whether female-female student-teacher matching is associated with higher math achievement, using data from 214 rural Chinese schools in China (N = 9,072). Rather than treating all estimators as mechanically identical replications of one coefficient, I use an identification-spectrum framework. Conservative within-school comparisons provide the most transparent but underpowered estimate (+0.077 SD), while broader graph-based, propensity-score, doubly robust, and machine-learning approaches rely more heavily on selection-on-observables but recover positive estimates of varying magnitudes. The disciplined estimates from OLS, DAG, IPW, and doubly robust methods range from +0.074 to +0.098 SD, with the conservative OLS estimate corresponding to roughly 37% of the baseline gender gap. Double machine learning produces larger estimates (+0.213 to +0.230 SD), which I interpret as flexible upper-range evidence rather than as a replacement for the conservative range.

Beyond average effects, the results suggest substantial heterogeneity. Causal forest estimates indicate that the gender-matching benefit is largest in low- and high-salary classrooms and close to zero in the middle of the salary distribution. Because teacher salary is a proxy for classroom and school conditions rather than a randomized policy lever, this pattern should be read as suggestive evidence about where matching may matter most, not as proof that salary itself causally moderates the effect. Overall, the results support a cautious conclusion: female-female math teacher matching is consistently positive across a range of identification strategies, but the magnitude depends on the baseline comparison and assumptions used. These findings suggest that teacher gender may be one school-level channel through which math gender gaps are either reinforced or narrowed.

Keywords: gender gaps, teacher-student matching, causal inference, heterogeneous treatment effects, STEM education

1 Introduction

Girls and boys enter kindergarten with statistically equivalent math scores (Fryer & Levitt 2010). By the time they finish primary school, that equivalence is gone, and a recent large-scale study finds that the divergence begins as early as first grade (Martinot et al. 2025). This suggests that the math gender gap is not only about measured ability; the school and social environment also matter. Families and schools in rural settings tend to allocate fewer educational resources to girls (Tsai et al. 2018). Girls internalize gender stereotypes through socialization in ways that discourage persistence in math (Keller et al. 2021), and because girls develop a comparative advantage in reading early on, gendered expectations can further steer capable students away from math-intensive paths (Breda & Napp 2019). Even subtle contextual cues matter: Steele & Ambady (2006) show that reminding female students of their gender, rather than their academic identity, measurably lowers their math performance. These effects can compound at the level of self-perception: exposure to gender stereotypes lowers girls' self-assessed math competence even when their actual scores are equivalent to boys', reducing their confidence in pursuing math-related choices (Correll 2001). The result is a gap that grows with schooling rather than disappearing.

One factor that may moderate how the environment shapes girls' math outcomes is teacher gender. Teachers hold gender-biased beliefs about math ability that systematically favor boys, and these biases feed back into student achievement (Terrier 2020). A female math teacher is less likely to exhibit such grading bias and may function as a role model who makes math feel more accessible to female students (Chan et al. 2021). Prior work finds evidence consistent with this, though the picture is mixed. Winters et al. (2013) documents a statistically significant relationship between female teachers and female students' achievement in middle and high school. Antecol et al. (2015), however, find a negative effect in disadvantaged US primary schools — driven specifically by female teachers with weaker math backgrounds — suggesting that teacher subject knowledge moderates the sign of the effect. The limitation shared by both studies is identification: teacher-student assignment is rarely random, and if the characteristics driving assignment are correlated with student outcomes, the estimated effect of teacher gender is confounded.

This paper uses data from a large-scale RCT conducted in rural China (Loyalka et al. 2019) to study whether female students taught by female math teachers have higher math achievement. The original experiment randomly assigned pay incentive designs across 216 schools in Shaanxi and Gansu provinces between 2012 and 2014 (N = 9,072 students). I extend the original study by examining a question the pay-design experiment was not designed to answer: whether female-female teacher-student matching is associated with higher standardized math scores. Because the pay-design randomization is orthogonal to teacher gender, it can be included as a control. The causal interpretation rests on the assumption that, conditional on baseline scores, individual characteristics, teacher covariates, and school and county fixed effects, teacher-student gender assignment is as good as random within the relevant comparison. Matched female-student–female-teacher pairs make up 18.6% of the sample, which provides meaningful raw variation, although the within-school identifying variation is much more limited.

I estimate the question through a sequence of regression, regularization, tree-based, graph-based, propensity-score, meta-learner, policy-tree, and double machine learning approaches. The range of methods serves two purposes: to test whether the female-female matching signal remains positive across different identifying assumptions and modeling choices, and to characterize treatment-effect heterogeneity that a single linear specification cannot recover.

A complication in this setting is that the female-female treatment cell can be compared against different baselines depending on the estimator. I therefore do not read the methods as mechanically identical replications of one coefficient. Instead, I use them as a sequence of related lenses around the same substantive question. The school fixed-effects OLS specification provides the most conservative and design-transparent comparison, but it is identified only from the small set of mixed-teacher schools. The DAG, IPW, and doubly robust estimators relax that restriction and use broader between-school comparisons under explicit adjustment assumptions. Double Machine Learning uses the richest control set and flexible nuisance-function estimation, returning larger estimates and the main heterogeneity evidence. The key robustness claim is therefore directional and comparative: as the identifying variation broadens and the modeling assumptions become stronger, the estimated female-female matching effect remains positive.

The estimates point in a consistent positive direction, but they should be read with different levels of caution. In the baseline OLS model, female students score 0.147 standard deviations lower than male students after conditioning on baseline scores, student demographics, teacher characteristics, and school fixed effects. The female-female matching coefficient is +0.077 standard deviations, which corresponds to approximately 37% of the baseline gap. Under this most conservative specification, school fixed effects absorb most between-school variation in teacher gender, so the school-clustered standard error is too large to reject the null at conventional levels. Methods that draw on broader between-school variation produce similar positive estimates: IPW gives +0.098 SD, a doubly robust estimator gives +0.087 SD, and the DAG backdoor gives +0.074 SD. These four disciplined estimates range from +0.074 to +0.098 SD. Double Machine Learning, using the richest control set and flexible residualization, produces larger cross-fitting-based estimates (+0.213 to +0.230 SD, all significant). I read the DML estimates as upper-range evidence under stronger modeling assumptions and as the basis for heterogeneity analysis, not as replacements for the conservative +0.07 to +0.10 SD range. The heterogeneity results suggest a non-monotonic pattern: teacher base salary appears to be the main effect modifier, with near-zero predicted effects in mid-salary classrooms and larger predicted effects at both the low and high salary extremes.

The remainder of the paper is organized as follows. Section 2 describes the data and presents summary statistics. Section 3 details each model specification and presents its results, moving from average effects to treatment effect heterogeneity. Section 4 discusses the findings, reconciles estimates across methods, and outlines policy implications. Section 5 concludes.

2 Data and Summary Statistics

2.1 Data Source

The data used in this study comes from a field experiment conducted in rural China. The original field experiment measured test scores in math, science, and reading to examine the impact of pay designs on teacher performance. The sample comes from two provinces: Shaanxi (ranked 12th in GDP per capita among China's 1st level administrative regions) and Gansu (ranked 27th). Among the two provinces, 216 schools were randomly selected from their nationally ranked "poverty counties", counties where annual earnings were below a certain threshold. All schools were publicly funded in rural China and had around 400 students. The data was collected through four surveys, conducted from 2012 to 2014, to explore different pay designs.

Student-level microdata was collected through two baseline surveys: one conducted in May 2012, and the other in September 2012. The surveys collected information such as age and gender, education level and occupation of parents, number of siblings, and family assets. Another endline survey (May 2014) was also conducted, focusing on non-cognitive factors (self-concept, anxiety, and intrinsic and instrumental motivation scales) towards math, curricular coverage and difficulty of math curriculum, time spent on math and other subjects per week, perceptions of teaching practices, care shown by teacher, management of the classroom, and communication by the teacher; as well as the involvement of parents in schoolwork.

Teacher-level microdata was collected in one baseline survey, conducted at the start of the school year (Sep. 2013). This survey collected data on the teacher's gender, ethnicity, age, experience level, and credentials, as well as attitude towards performance pay alongside their current performance pay. The teacher's perceived returns on teaching effort for individual students within the class were also collected. At the end of sixth grade, a nearly identical survey was given to teachers.

During the 3 surveys issued to students, a 35-minute standardized math test was conducted. These tests were constructed by trained psychometricians in order to make sure a fair measurement was taken. The items on the math test were selected from the standardized math curricula for primary school students in Shaanxi and Gansu. The content validity was then verified by multiple experts, and then the tests were validated to ensure knowledge tested was evenly distributed (i.e., no bottom or top coding). The test scores were normalized against a control group, with each test's scores being normalized separately, expressing the estimated effects in standard deviations. In this study, the entire original sample was used (N = 9072).

2.2 Key Variables and Summary Statistics

Table 1 reports summary statistics for the key variables. All math scores are standardized using the mean and standard deviation of the control group. Despite being drawn from nationally designated poor counties, there is meaningful variation in family background: 53% of fathers and 33.6% of mothers completed junior high school, and girls on average came from slightly more disadvantaged households than boys — a pattern consistent with the broader literature on gender and resource allocation in rural China (Tsai et al. 2018).

Female math teachers are not rare in the sample — they teach 46% of students — but the joint cell of interest is a minority: female students matched with a female math teacher account for only 18.6% of the full sample (21.5% of the estimation sample). This matched cell provides the key source of variation for our treatment of interest; the corresponding shares are reported in Table 1.

All conditioning variables are measured pre-treatment: the two baseline scores and student characteristics come from the 2012 surveys, and teacher characteristics from the September 2013 survey administered at the start of the sixth-grade year — before the grade-6 math teacher, whose effect the May 2014 endline measures, had taught these students.

Table 1: Summary Statistics

	Count	Mean	SD	Min	Max
Final score	8,559	0.07	1.047	−2.715	2.8
Math baseline 1	7,996	−0.024	1.013	−3.27	2.025
Math baseline 2	8,136	0.014	1.027	−3.21	2.039
Female	7,996	0.488	0.500	0	1
Female math teacher	9,006	0.457	0.498	0	1
Female student × female teacher	9,006	0.186	0.389	0	1
Male student × male teacher	9,006	0.254	0.435	0	1
Student age	7,992	12.027	1.033	9	19
SES index	7,996	−0.602	1.023	−2.168	4.097
Class size	9,072	44.055	15.723	7	103
Teacher experience (yrs)	9,006	12.339	9.895	0.083	54
Teacher base pay (yuan)	9,006	2,935.6	915.1	—	5,220

Note: Teacher base pay minimum excludes values of zero, which are concentrated in five schools and reflect non-response rather than true zero salaries; these observations are recoded as missing prior to analysis.

2.3 Balance Check

Table 2 reports baseline student characteristics separately for students taught by female versus male math teachers, and Figure 1 shows the corresponding raw distributions. Raw comparisons suggest that students with female teachers have higher baseline math scores (differences of 0.278 and 0.217 standard deviations for the two baseline waves) and come from slightly more advantaged family backgrounds. However, these differences reflect between-school sorting rather than within-school selection: once school fixed effects are absorbed, none of the six baseline characteristics significantly predict teacher gender within schools. A joint F-test confirms that all six characteristics are jointly insignificant ($F = 0.97$, $p = 0.44$), supporting the conditional independence assumption underlying our identification strategy.

Table 2: Balance Check: Baseline Characteristics by Teacher Gender

	Male teacher	Female teacher	Difference	Within-school p
Math baseline 1	−0.147	0.131	0.278	0.100
Math baseline 2	−0.085	0.132	0.217	0.398
Student age	12.066	11.974	−0.092	0.408
SES index	−0.646	−0.549	0.097	0.168
Father's education (JHS+)	0.501	0.566	0.065	0.899
Mother's education (JHS+)	0.319	0.362	0.043	0.991
Joint F-test (within-school)				$p = 0.44$

Note: The mean columns are computed characteristic-by-characteristic on all non-missing observations, so each row uses every student for whom that variable is recorded (between 7,867 and 8,072 students, depending on the characteristic). The within-school p-values and the joint F-test, by contrast, require a single complete-case sample in which all six characteristics are non-missing (N = 7,744); the two samples differ by students missing at least one baseline variable, which shifts the reported means by at most 0.007. Within-school p-values come from OLS regressions of teacher gender on each characteristic with school fixed effects and HC1 robust standard errors; the joint F-test uses a single regression with all six characteristics and school fixed effects, and tests only the baseline characteristics, not the school dummies. Appendix Table 9 reports the fully complete-case version of this table — means, standardized mean differences, within-school coefficients, and p-values all computed on the same 7,744 students; the substantive conclusion is unchanged.

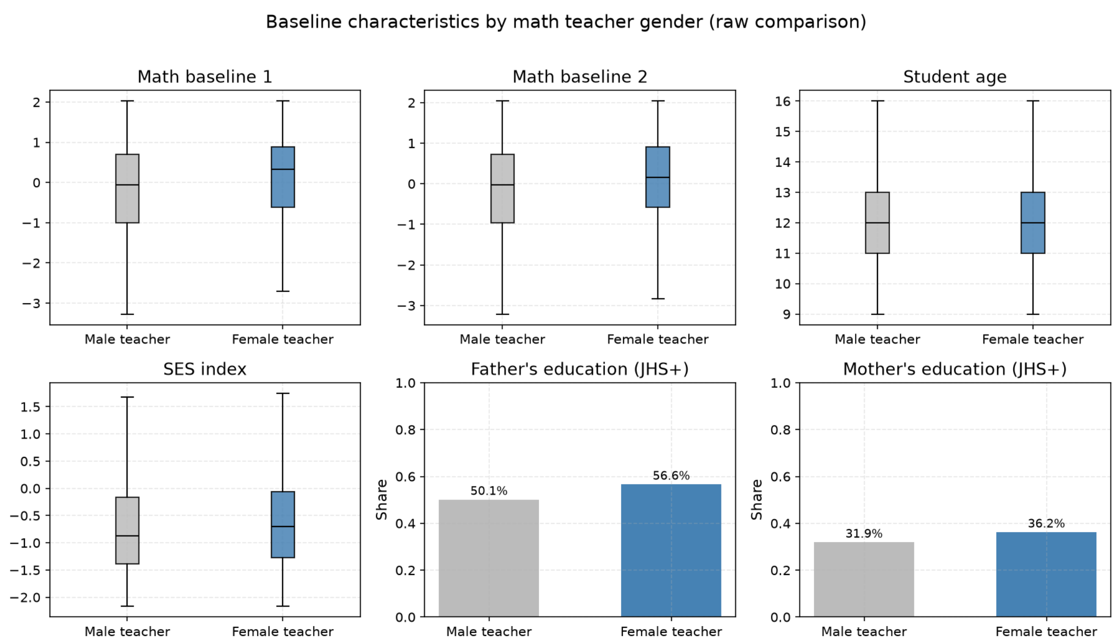


Figure 1: Baseline student characteristics by math teacher gender (raw comparison, before school fixed effects). The visible advantages of female-teacher students reflect between-school sorting; Table 2 shows they disappear within schools.

3 Model Analysis and Implementation

Before presenting the models, it is useful to clarify how the estimates should be read. The female-female treatment cell is mechanically built from student gender and teacher gender, and teacher gender is nearly constant within most schools. As a result, different estimators compare the female-female cell against different baselines. I use these methods as an identification spectrum rather than as exact replications of one coefficient. The fixed-effects OLS specification is the most conservative lens: it is closest to the female-student contrast between female and male math teachers, but it relies only on the 12 schools where both male and female math teachers are observed. The DAG, IPW, and doubly robust estimators use broader comparison bases under explicit adjustment assumptions, gaining precision at the cost of stronger selection-on-observables assumptions. DML then uses the richest controls and flexible, regularized nuisance estimation, which makes it useful for heterogeneity analysis but also the most model-dependent average estimate. The robustness claim is therefore that closely related contrasts around the same substantive question point in the same positive direction.

3.1 Linear Regressions

Model Specification As the first step I created the following linear regression model to capture the gender gap in math, while capturing the impact of gender-matching teachers on students.

$$y_{ijc} = \alpha + Female_{ijc}\beta + T'^{t}_{ijc}\phi + X'_{ijc}\gamma_i + \tau'_c + \epsilon_{ijc}(2)$$

where, y_{ijc} is the *standardized math test score* for individual student i at school j in county c. $Female_{ijc}$ is a dummy variable if the individual i at school j in county c is female. T'^{t}_{ijc} is a set of dummy variables for additional interests – the impact of gender matching teacher on students, where t = {Female Student Female Teacher, Male Student Male Teacher}. $X'_{ijc}$is a vector of control variables, including individual-specific data (two waves of baseline achievement scores and student age, gender, parent educational attainment, a household asset index, class size, teacher experience, and teacher base salary) The controls also include the randomized treatment assignments from the original study (Loyalka et al. 2019); because those assignments are not the focus of this paper, their coefficients are not reported. τ'_c denotes county and school fixed effects.

Result The model confirms a persistent gender gap across both specifications. In Column 1, girls score 0.147 standard deviations lower than boys after conditioning on two waves of baseline scores, student demographics, teacher characteristics, and school fixed effects.

Column 2 adds the gender-matching interactions. Girls without a gender-matched teacher score 0.209 standard deviations lower than boys without a gender-matched teacher — the gap is larger than Column 1 because the baseline now compares opposite-gender configurations, isolating what gender matching contributes on top. The gender-matching coefficient is +0.077 standard deviations, directionally consistent with the hypothesis and closing roughly 37% of the baseline gap. The standard error is large, however: school fixed effects absorb most of the between-school variation in teacher gender, leaving identification to the small share of schools where male and female teachers coexist within the same grade — only 12 of the 214 schools, containing 475 of the 7,373 students in the estimation sample. This is not merely a power issue but an algebraic one: in the remaining 202 schools all math teachers share the same gender, making teacher gender a school-level constant, so the two matching indicators and the female dummy are perfectly collinear with the school dummies (f_same_sex – m_same_sex – female is constant within each such school). Separating the matching effect from the baseline gender gap therefore rests entirely on those 12 mixed schools. With school-level clustering, the estimate does not reach conventional significance thresholds ($p = 0.43$). The coefficient on boys with male teachers is -0.051, also not significant. The imprecision is a power problem driven by limited within-school identifying variation — not evidence of a null effect. This specification is therefore the most design-transparent estimate in the paper, but also the most statistically underpowered. The following estimators broaden the comparison base and should be read as less conservative but more precise lenses on the same substantive question.

Table 3: Effect of Teacher-Student Gender Match on Math Test Scores

	Outcome Variable: standardized math test score	
	(1) Base	(2) Gender-Matching
Female	-0.147***	-0.209*
	(0.019)	(0.099)
Female student female teacher		0.077
		(0.097)
Male student male teacher		-0.051
		(0.098)
Observations	7373	7373
R^2	0.476	0.477
Adjusted R^2	0.474	0.475
Residual Std. Error	0.756 (df = 7343)	0.756 (df = 7341)
F Statistic	137.696*** (df = 29; 7343)	130.007*** (df = 31; 7341)

Note: The standard errors are clustered at the school level *p<0.1; **p<0.05; ***p<0.01

3.2 Linear Regression with Regularization

Ridge and LASSO regression are included not as causal estimators but as prediction-based checks on whether the main controls carry predictive signal. Because regularization penalizes large coefficients, variables that remain active near the cross-validated optimal λ provide descriptive evidence about prediction, not causal importance. The regression specification mirrors the OLS models; the objective functions are listed below.

3.2.1 Ridge Regression

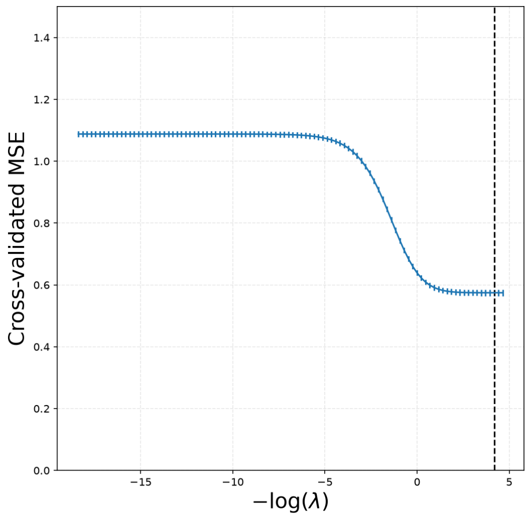
Ridge Regression applies penalties to coefficients, resulting in coefficient shrinkage. The shrinkage is proportional, so the exercise is useful for checking predictive stability across the control set.

Objective Function: The shrinkage is achieved by optimizing the following objective function, where the penalty terms has a l_2 form.

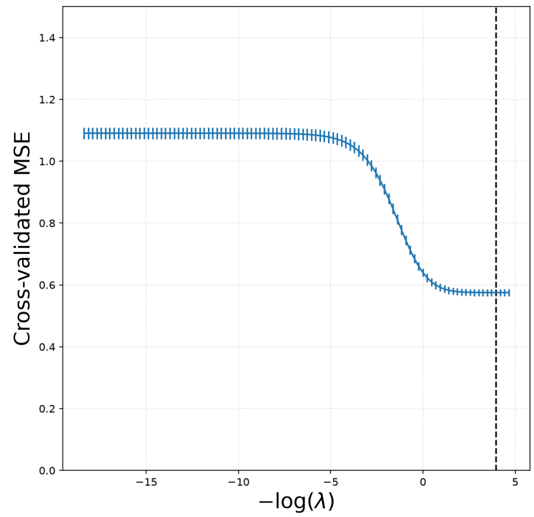
$$\min_{b \in \mathbb{R}^p} \sum_i (y_i - b' X_i)^2 + \lambda \cdot \sum_{j=1}^{p} b_j^2 \hat{\psi}_j$$

Result The optimal penalty λ^* is selected by five-fold cross-validation, choosing the value that minimizes held-out Mean Squared Error. The cross-validated optimum is $\lambda^* = 0.0152$ for both models (shown by the vertical dashed line). This value is small: penalizing coefficients beyond this point increases test MSE, so the cross-validation exercise gives no strong predictive reason to shrink the main controls further. This is a prediction check only; it does not establish that each variable is causally important.

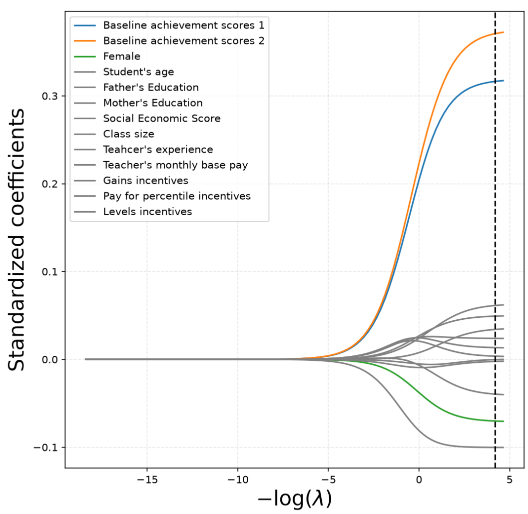
Figure 2: Ridge: regularization effect on model performance



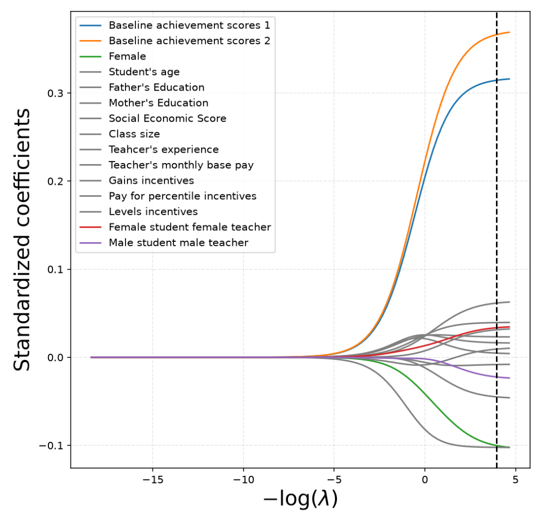
(a) Five-fold cross-validation MSE for the Base model



(b) Five-fold cross-validation MSE for the Gender-Matching model



(c) Penalized Ridge coefficients as a function of $-\log(\lambda)$: Base model



(d) Penalized Ridge coefficients as a function of $-\log(\lambda)$: Gender-Matching model

Note: School and county dummies are excluded from the above graphs for simplicity. The raw graphs are attached in the appendix

3.2.2 LASSO Regression

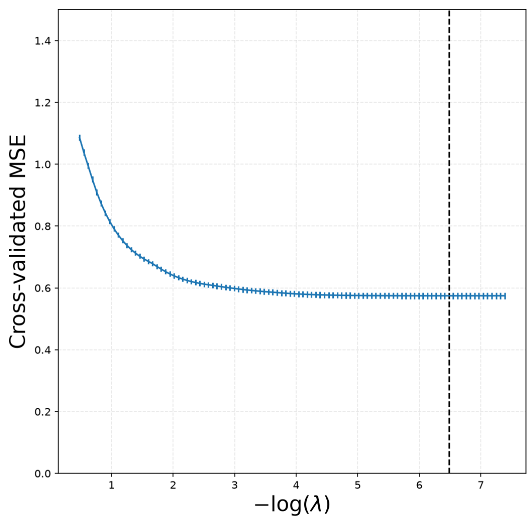
Unlike Ridge, LASSO Regression applies penalties to coefficients disproportionally, allowing coefficients to drop out.

Objective Function The shrinkage is achieved by optimizing the following objective function, where the penalty terms has a l_1 form.

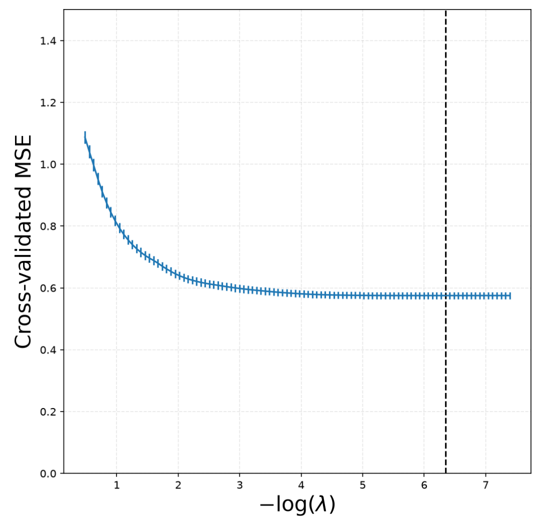
$$\min_{b \in \mathbb{R}^p} \sum_i (y_i - b'X_i)^2 + \lambda \cdot \sum_{j=1}^{p} |b_j| \hat{\psi}_j$$

Result The cross-validated optimum is $\lambda^* = 0.0016$ (shown by the vertical dashed line). As with Ridge, this small value gives no strong predictive reason to remove the main controls from the specification. The exception is informative: in the Gender-Matching model, the coefficient on *male student with male teacher* (m_same_sex) nearly reaches zero at λ^*, indicating it contributes little to prediction in this setup. This is consistent with the OLS result, where m_same_sex is small and statistically insignificant. The variable is nonetheless retained in subsequent models for completeness and to keep the female gender-matching coefficient interpretable against the full set of gender cells.

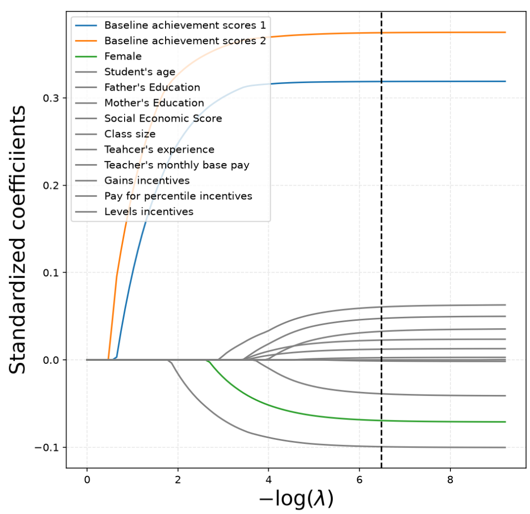
Figure 3: LASSO: regularization effect on model performance



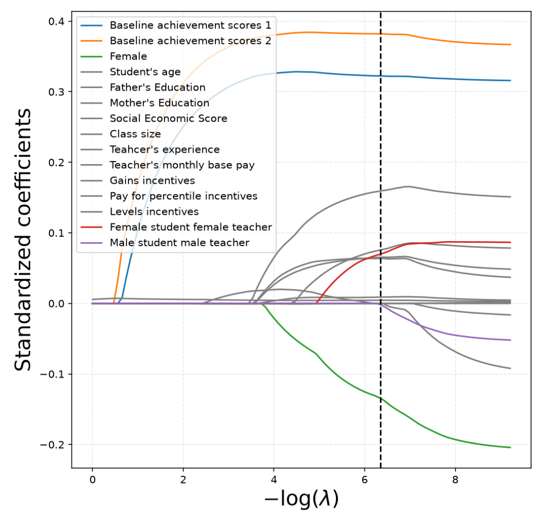
(a) Five-fold cross-validation MSE for the Base model



(b) Five-fold cross-validation MSE for the Gender-Matching model



(c) Penalized LASSO coefficients as a function of $-\log(\lambda)$: Base model



(d) Penalized LASSO coefficients as a function of $-\log(\lambda)$: Gender-Matching model

Note: School and county dummies are excluded from the above graphs for simplicity. The raw graphs are attached in the appendix

3.3 Ensemble Trees Method

The estimators so far all impose a linear functional form. Tree-based models relax that restriction by allowing nonlinear splits without specifying them in advance. As with Ridge and LASSO, the trees in this section are prediction models rather than causal estimators; their role is to describe predictive structure and possible nonlinearities before the explicitly causal methods that follow.

3.3.1 Decision Tree

As shown in Figure 4, the tree's root node splits on baseline achievement scores. This suggests that prior achievement carries the strongest predictive information for final achievement, which is expected in a standardized-test setting. Besides the academic factor, socioeconomic status is a key splitting criterion on several branches, showing that the model can detect disparities linked to family background. Teacher experience, salary, and class size also appear, although they are further down the tree. The fact that school ID appears as a splitting criterion is also consistent with school-level heterogeneity.

Looking at the extreme branches of the tree, high baseline achievement, socioeconomic status, and favorable learning conditions (smaller classes, well-paid teachers) lead to the highest prediction branch. This matches research showing that prior achievement and access to resources are strongly associated with educational outcomes. Conversely, the opposite characteristics lead to the lowest prediction branch, reinforcing the descriptive importance of systemic inequality for disadvantaged students. These branches should not be read causally: the tree identifies where low predicted achievement concentrates, not which intervention would raise scores. Its role is to map predictive structure before the causal estimators that follow.

The Feature Importance Index shows baseline achievement scores 1 and 2 to be the strongest predictors in this model. Socioeconomic status, teacher base monthly salary, and class size are the next strongest predictors, although their feature-importance values are much smaller.

In terms of prediction, the boosting model has the lowest Mean Squared Error among the three tree-based models. For that reason, later machine-learning specifications use gradient boosting rather than the single regression tree or bagging estimator.

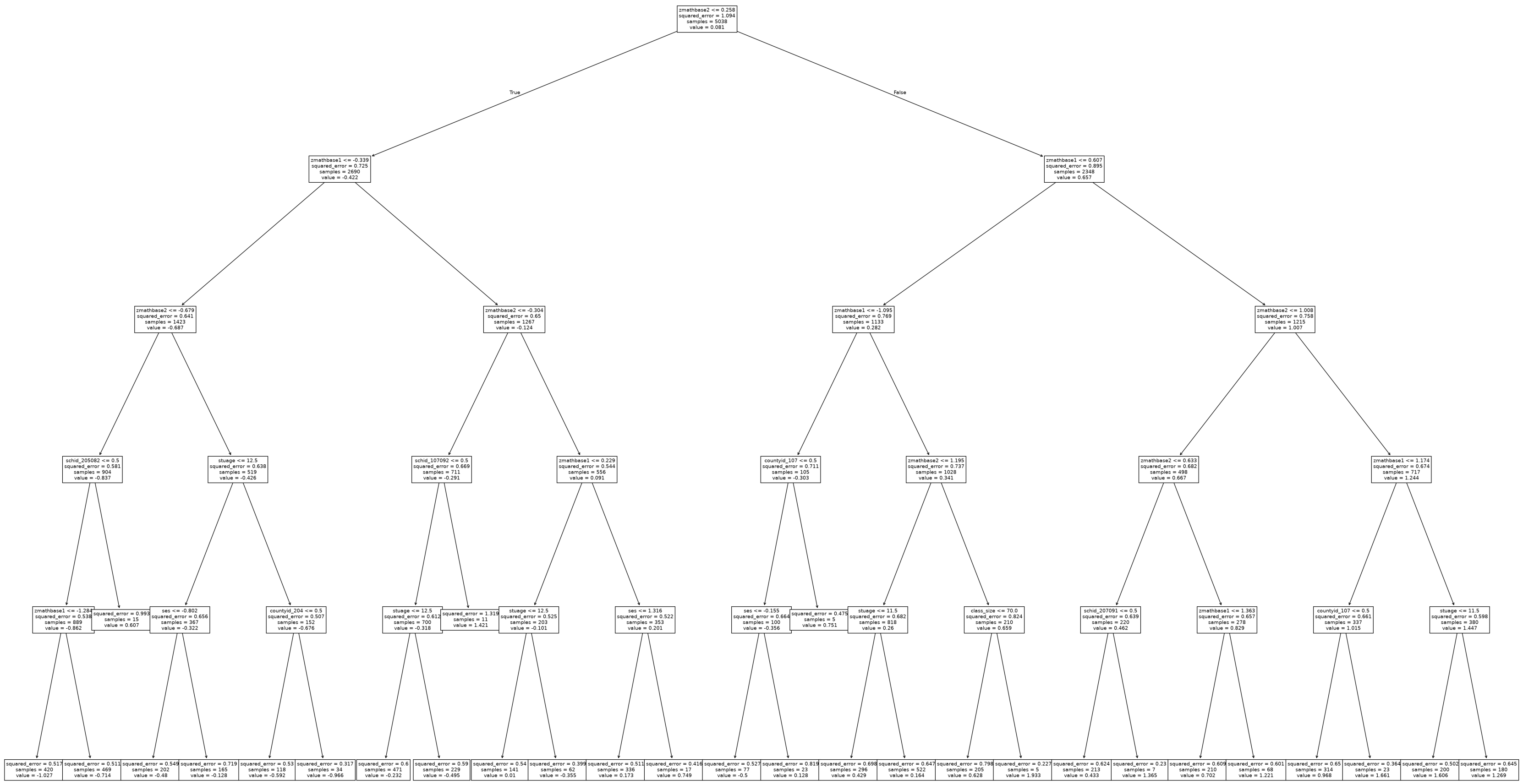


Figure 4: Regression Tree

	Importance
Baseline achievement score 2	0.409373
Baseline achievement score 1	0.154122
Social economic score	0.055577
Teacher's base monthly salary	0.039246
Class size	0.037650
Teacher's experience	0.034000
Student age	0.028833
School ID 205082	0.014652
School ID 107081	0.007588
Female	0.007333

Table 4: Feature importance index (only top 10 included)

Table 5: Tuned MSE for Different Models

Model	Tuned MSE
Regression Tree	0.655
Bagging	0.565
Boosting	0.535

Mean Squared Error (MSE) Comparison

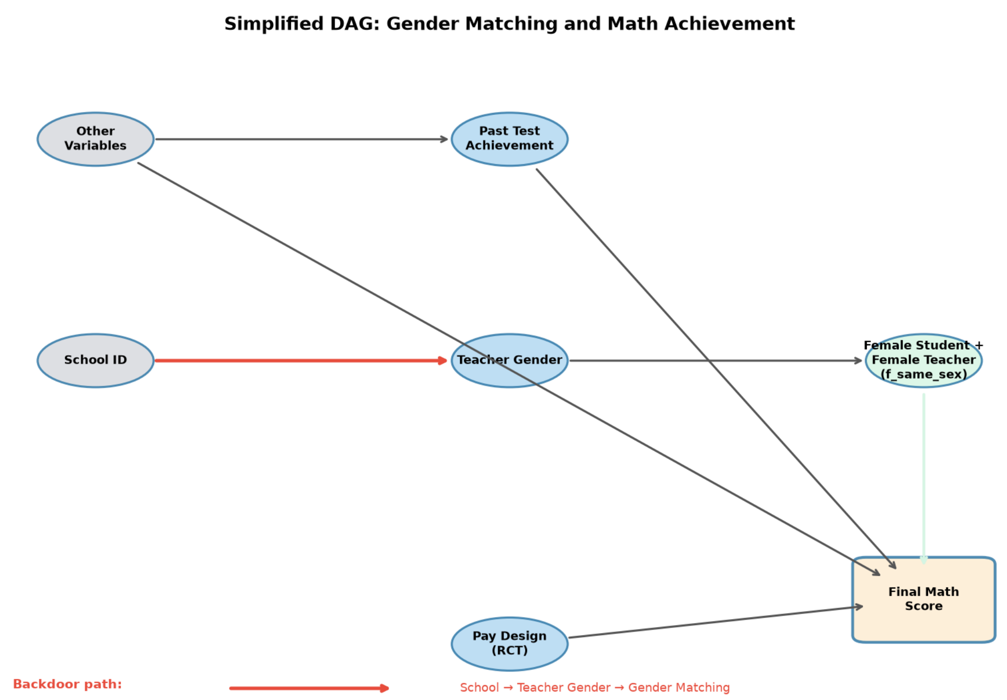
3.4 Directed Acyclic Graphs

The trees rank variables by predictive power, but prediction is not identification: knowing that baseline scores predict outcomes does not say which variables must be adjusted for before the gender-matching coefficient can carry a causal reading. Directed acyclic graphs (DAGs) make that question explicit: write down the assumed causal structure, and let the graph identify the minimal set of variables that must be adjusted for. In this session, only one DAG graph is constructed, which captures both the gender gap and the gender-matching effects.

Identification Strategy

The important assumption for the DAG approach is unconfoundedness: there should be no unobserved confounder affecting both the treatment and the outcome after the graph-defined adjustment set is applied. I estimated three separate models corresponding to three treatments: female, female-student–female-teacher, and male-student–male-teacher. As shown in Figure 5, there is a backdoor path between each treatment and the outcome. Under the assumed graph, causal interpretation is possible if the relevant backdoor paths are blocked. I group the observed covariates into four main areas: county-school fixed effects, student ability, family investment, and classroom quality. County-school fixed effects include the school and county identifiers. Ability is proxied by baseline achievement and age. Family investment is proxied by father's education, mother's education, and family socioeconomic status. Classroom quality is proxied by teacher experience, teacher monthly base salary, and class size. These are imperfect proxies for broader constructs, but they are all measured before the outcome: student and family variables come from the 2012 surveys, and teacher and classroom variables come from the September 2013 survey at the start of the school year. This timing is why the graph contains no arrows from gender matching into the baseline scores: the grade-6 math teacher had not yet met these students when the baselines were taken. The pay-design group assignment from the original experiment is randomized and therefore orthogonal to teacher gender; in the causal graph it points only at the outcome, so it is not part of the backdoor adjustment set and enters the models as a control to improve precision rather than for identification. Figure 5 is simplified for readability; the original DAG is attached in the appendix.

Figure 5: Simplified DAG



Result Under the DAG framework, conditioning on the backdoor adjustment set identified by the causal graph gives a gender-matching estimate of +0.074 SD. It is worth being precise about what the graph adjusts for. Although school and county dummies are present in the data passed to the estimator, the backdoor criterion identifies teacher gender and student gender as the *minimal sufficient* adjustment set under the assumed graph: once teacher gender is conditioned on, the graph treats the path from school membership into the treatment (`school` → `t_female` → `f_same_sex`) as blocked, so the school dummies are redundant and the estimator does not use them. The result is therefore a broad between-school comparison, not a within-school one. This matters for interpreting the number: it sits close to the fixed-effects OLS estimate, yet the two arrive there by different routes. OLS leans entirely on the 12 mixed-teacher schools, while the backdoor estimator compares students across all schools after a minimal gender-based adjustment. I do not read the DAG estimate as more correct than OLS; I read it as evidence that a different graphical lens gives the same positive direction. The next two sections test broader comparisons more directly, by modeling the treatment rather than the outcome.

	Estimate
Female	-0.159
Female student Female teacher	0.074
Male student Male teacher	-0.038

Table 6: DAG backdoor linear regression estimate

3.5 Propensity-Score Methods: Inverse Weighting and Doubly Robust Estimation

OLS and the DAG estimator both model the outcome. The next two estimators model the *treatment* instead, through the propensity score $e(X) = P(T = 1 \mid X)$ — the probability that a student is gender-matched given covariates. Inverse propensity weighting (IPW) reweights the sample by the inverse propensity; the doubly robust estimator (DR, also called augmented IPW) adds an outcome model on top of the same propensity score, so that it stays consistent if *either* model is correctly specified. Because both rest on the same $e(X)$ and the same identifying assumptions, they are presented together.

3.5.1 Inverse Propensity Weighting (IPW)

Model Specification

IPW reweights the sample so that treated and untreated students are comparable by construction.

The propensity score is estimated with an L2-regularized logistic regression ($C = 1$, SAGA solver) of the gender-matching indicator on baseline student and teacher characteristics plus school and county fixed effects. One modeling choice shapes the estimand: the propensity model omits student sex and the male–male matching cell, departing from the full gender-cell control set used in OLS. This deliberate choice widens the identifying variation beyond the 12 mixed-teacher schools, but it also asks a broader question. Rather than isolating the female-female cell against the female-student/male-teacher baseline, IPW estimates whether the gender-match signal persists in a reweighted full sample. The choice is therefore a stress test of robustness, not a claim that the IPW coefficient is directly comparable to the OLS cell coefficient. Each observation is then weighted by $T/e(X)$ or $(1-T)/(1-e(X))$, and the ATE is the weighted mean outcome difference between the two groups.

Identification Strategy

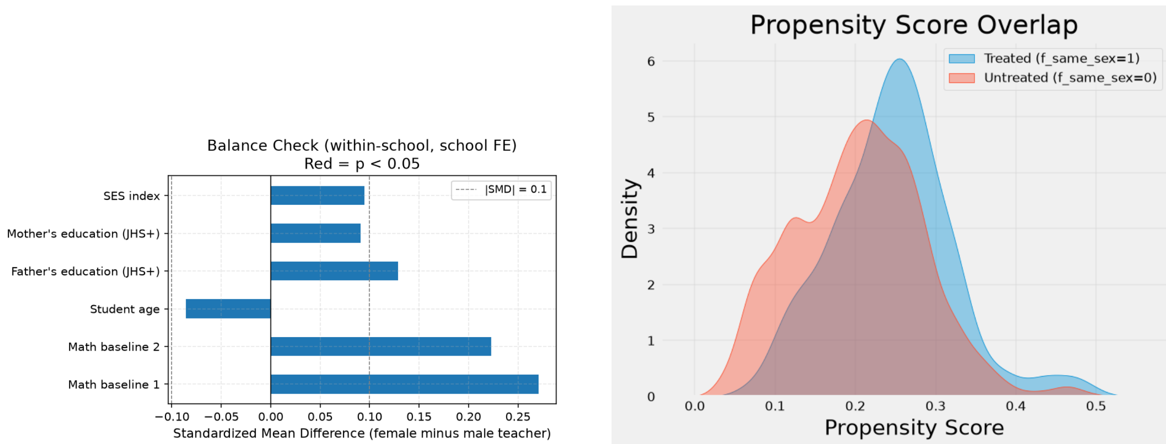
IPW requires two assumptions. The first is the conditional independence assumption (CIA):

$$Y_1, Y_0 \perp T \mid X$$

Because teacher-student gender matching was not randomized, CIA cannot be guaranteed by design. The balance check in Section 2.3 provides supportive evidence for the conservative within-school comparison: conditional on school fixed effects, the six baseline student characteristics are jointly insignificant predictors of teacher gender ($F = 0.97$, $p = 0.44$). IPW goes beyond that within-school comparison by using broader between-school variation, so its causal interpretation requires the stronger assumption that the observed pre-treatment covariates and fixed effects absorb the relevant selection across schools. Figure 6a visualizes the corresponding standardized mean differences. All covariates entering the propensity model are pre-treatment — student variables from the 2012 surveys, teacher variables from the September 2013 survey at the start of the school year — so none of them can be a consequence of the matching itself.

The second assumption is positivity (overlap): every student must have a positive probability of receiving either teacher gender, $0 < e(X_i) < 1$. Figure 6b shows the estimated propensity score distributions for treated and untreated students. The distributions overlap across most of the support — both groups fall between roughly 0 and 0.5, with no mass at the extremes — suggesting that the comparison is not extrapolating into regions with no counterparts and that no weights explode. The regularization is doing real work here: within a school, treatment is a deterministic function of student and teacher gender, so an unpenalized logistic regression with school dummies would push the propensity to the boundary in single-teacher-gender schools. The L2 penalty stabilizes the propensity model and prevents separation; it does not itself create identification. Note that overlap is a necessary condition for identification but does not itself establish CIA — the two assumptions are distinct.

The ATE estimate is +0.098 SD (95% CI: 0.075–0.172) for the reweighted female-female matching signal — consistent in direction with OLS (+0.077) and the DAG backdoor (+0.074), and the first estimate in the paper whose confidence interval excludes zero. Freed from the fixed-effects algebra of Section 3.1, the reweighted between-school comparison has enough variation to be precise; the price is that it leans harder on CIA than the within-school design does.



(a) Within-school covariate balance

(b) Propensity score overlap

Figure 6: IPW diagnostics: standardized mean differences in baseline covariates by teacher gender (left) and estimated propensity score distributions for treated and untreated students (right).

3.5.2 Doubly Robust Estimation (AIPW)

Identification Strategy

The doubly robust (DR) estimator augments the IPW step with a linear outcome model — the "augmented IPW" construction. It is consistent if *either* the propensity model or the outcome model is correctly specified — hence "doubly robust." Under the potential-outcomes framework, the estimator takes the form:

$$\hat{\tau}^{\mathrm{DR}} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{T_i(Y_i - \hat{\mu}_1(X_i))}{\hat{e}(X_i)} + \hat{\mu}_1(X_i)\right] - \frac{1}{n}\sum_{i=1}^{n}\left[\frac{(1-T_i)(Y_i - \hat{\mu}_0(X_i))}{1-\hat{e}(X_i)} + \hat{\mu}_0(X_i)\right]$$

where $\hat{e}(X_i)$ is the estimated propensity score from the IPW step and $\hat{\mu}_1(X_i)$, $\hat{\mu}_0(X_i)$ are OLS-predicted potential outcomes fitted separately on the treated and untreated sub-samples. The same propensity model is reused here — logistic regression with L2 regularization ($C = 1$), school and county fixed effects, and the SAGA solver — so the DR estimate inherits the same overlap and regularization properties as the IPW estimate. The identifying assumptions are the same as for IPW: CIA and positivity.

Result

The DR ATE estimate is +0.087 SD for the same reweighted female-female matching signal. This sits between the IPW estimate (+0.098 SD) and the OLS/DAG estimates (+0.074–+0.077 SD), which is the expected pattern: DR averages information from both the propensity and the outcome surface, so when the two components pull in slightly different directions, the combined estimate lands in between. The bootstrap confidence interval is wide (95% CI: −0.26, +0.44), reflecting high variance in the OLS outcome model when fitted to 200+ school dummies across many bootstrap samples. The point estimate is directionally informative, but this specification does not independently deliver precision.

3.6 Meta Learners

Every method up to this point has reported a single average effect for everyone, and the conservative estimates are positive in the +0.074 to +0.098 SD range, with IPW significant. The remaining sections — meta-learners, a policy tree, and double machine learning — shift the question from *whether* gender matching helps to *whom* it helps most. The meta-learner estimates are implemented within the DAG framework — they re-fit the same backdoor adjustment set (teacher and student gender) with flexible ML base learners in place of the linear model, reporting one number per treatment arm. One caveat belongs up front: unlike the propensity-score and DML estimators, meta-learners do not come with valid confidence intervals, so the estimates here are read for sign and rough magnitude rather than precision, for reasons taken up after the table. Estimated on the full sample, the tuned T-learner gives +0.085 SD, inside the conservative range, while the X-learner with a regularized ($C = 1$) propensity gives a smaller but still positive +0.042 SD. The S-learner, which pools the treatment into a single outcome model, collapses to essentially zero (−0.004 SD) for reasons taken up after the table. These averages are all estimated on the full sample, and the useful meta-learner evidence is directional rather than definitive. The distinct question of *whom* the effect concentrates on is left to the policy tree and DML sections, which are built for it.

The three learners differ only in how they turn base regressions into a CATE. Writing $\mu_t(x) = E[Y \mid X = x, T = t]$ for the outcome model in arm t and $e(x) = P(T = 1 \mid X = x)$ for the propensity score, the S-learner fits one pooled model with the treatment as a feature and differences it,

$$\hat{\tau}_S(x) = \hat{\mu}(x, 1) - \hat{\mu}(x, 0);$$

the T-learner fits the two arms separately and differences them,

$$\hat{\tau}_T(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x);$$

and the X-learner adds a second stage, imputing each unit's effect against the opposite arm's model — $\tilde{D}_i = Y_i - \hat{\mu}_0(X_i)$ for treated units, $\tilde{D}_i = \hat{\mu}_1(X_i) - Y_i$ for controls — regressing $\hat{\tau}_1, \hat{\tau}_0$ on those imputed effects, and blending them by propensity,

$$\hat{\tau}_X(x) = e(x)\,\hat{\tau}_0(x) + \big(1 - e(x)\big)\hat{\tau}_1(x).$$

The S-learner's single pooled model is exactly why a treatment that is a weak predictor of the outcome can be regularized away; keeping the arms separate, as the T- and X-learners do, avoids that failure mode.

The tuned T-learner (+0.085 SD) lands inside the +0.074 to +0.098 SD band of the regression and reweighting methods; the X-learner (+0.042 SD) is smaller but remains positive. The S-learner is the lone estimate at essentially zero (−0.004 SD). That dissent is the textbook S-learner failure rather than genuine disagreement: with the treatment folded into a single outcome model alongside roughly 180 other features, a cross-validated learner regularizes away an effect an order of magnitude smaller than the baseline-score signal. The X-learner estimate, by contrast, is robust to how its propensity model is fit. Removing the $C = 1$ penalty drives the estimated propensities to the boundary and inflates the implied weights by orders of magnitude — the same school-dummy separation that motivates the penalized propensity throughout the paper — yet the point estimate barely moves (+0.042 → +0.053 SD), because the final estimate is a convex combination of two same-signed effect models rather than a raw reweighting.[1]

Two things temper how much weight these point estimates can carry. First, meta-learners impose no structure on how their base learners trade off bias against variance, and as a result valid variance

[1] An off-the-shelf X-learner (EconML) run with an unregularized propensity returns −0.060 SD on the same data. Because my own four-stage X-learner does *not* reverse sign when its propensity is left unregularized, I read that negative value as an artifact of how that implementation handles a degenerate propensity rather than as a substantive result, and do not report it in Table 7.

Learner (`f_same_sex`)	**ATE (SD)**	**Note**
S-learner	-0.004	Weak treatment diluted in a single pooled model
T-learner (tuned)	$+0.085$	Inside the conservative range
X-learner	$+0.042$	Positive, but smaller than the range

Table 7: Full-sample meta-learner estimates for female gender matching. No specification carries a valid confidence interval. The T-learner lands inside the +0.074 to +0.098 SD range of the regression and reweighting methods; the X-learner is smaller but positive; the S-learner's pooled model shrinks the weak treatment toward zero.

estimation for S-, T-, and X-learners remains an open problem: none of the numbers in Table 7 carries a confidence interval, and bootstrapping them would not resolve the underlying issue. This is the sense in which the causal forest of Section 3.8 is on firmer ground — it has an asymptotically Gaussian sampling distribution and therefore genuine intervals, which the meta-learners lack. Second, the one specification that dissents — the S-learner, at essentially zero — does so for a well-understood reason, the weak-treatment shrinkage of a pooled model, rather than because the learners genuinely disagree about the effect; the two that keep the treatment arms separate remain positive and fall in the same broad order of magnitude as the disciplined methods. Read together with the absent confidence intervals, this means the meta-learners corroborate the sign and rough magnitude of the effect through their well-specified variants but cannot sharpen it. The one method that captures this heterogeneity *and* delivers valid confidence intervals is the causal forest inside the DML framework, to which the analysis turns after the policy tree.

3.7 Policy Tree

The meta-learners assign each student an estimated CATE; a policy tree compresses those estimates into a simple rule that highlights where the predicted matching gains are largest. The policy tree uses Conditional Average Treatment Effects (CATE) by splitting based on a feature's ability to explain heterogeneity. The first split in the policy tree occurs when the teacher's base salary is less than or equal to 3362, indicating that salary is the strongest observed divider of treatment-effect heterogeneity. Subsequent splits are secondary, suggesting salary captures primary heterogeneity. The tree alone only locates the split, not the shape of the relationship; the DML section below traces the full CATE-by-salary curve, and it turns out not to be monotonic. A teacher's salary likely proxies for the teacher's own experience, school resources, and classroom conditions. It is therefore useful for targeting and interpretation, but this analysis does not identify salary itself as a causal lever. Splitting on a teacher characteristic is legitimate here because it is pre-treatment: base salary was recorded in the September 2013 baseline survey, before the treatment year, and the experimental pay incentives added bonuses on top of base salary rather than changing it.

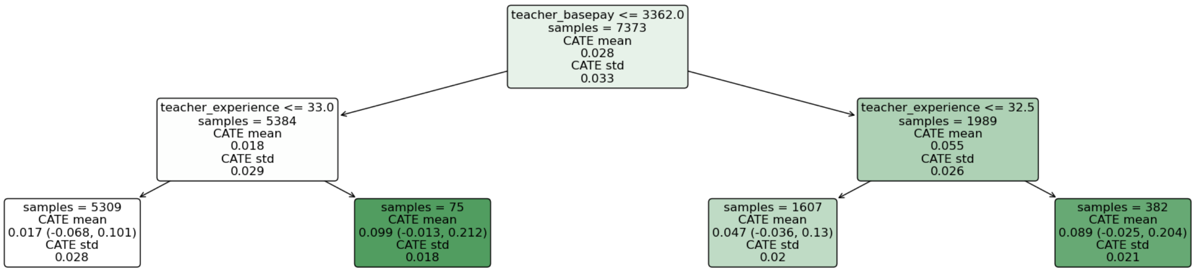


Figure 7: Policy Tree: Identifying Key Predictors of Treatment Effect Heterogeneity

3.8 Double Machine Learning

The meta-learners suggested that the treatment effect may be heterogeneous, and the policy tree pointed at a teacher characteristic, base salary, as the main axis. This section examines both patterns with double machine learning (DML), which delivers larger average estimates and the main heterogeneity evidence.

Identification Strategy

Double machine learning, also known as debiased machine learning, uses orthogonalization to reduce regularization bias in the final treatment-effect estimate. Both the outcome and the treatment are first predicted from the covariates with flexible learners under five-fold cross-fitting, and the causal coefficient comes from regressing the outcome residuals on the treatment residuals. The first-stage treatment model plays the role the propensity score played in IPW, so the same caveat carries over: treatment is not randomly assigned, and the key assumption of unconfoundedness is still required,

$$Y_1, Y_0 \perp T \mid X, W.$$

As in the earlier sections, both the controls W and the effect modifiers X are measured pre-treatment — student variables in the 2012 surveys, teacher variables in the September 2013 survey — so the residualization never conditions on post-treatment variables.

One difference from the IPW specification deserves a note, because it is exactly the issue raised by the fixed-effects OLS. Here the control set W is built for maximal confounding control and therefore includes the student's own sex, the male–male matching dummy, and the school and county dummies all at once — the same three ingredients that, combined, make the gender-matching indicators collinear within single-teacher schools. Had the first stage been an ordinary linear projection, the treatment residual $\tilde{T} = T - \hat{E}[T \mid X, W]$ would have collapsed toward zero in the 202 single-teacher schools, and identification would have fallen back onto the 12 mixed-teacher schools exactly as in Section 3.1. What prevents collapse is the regularized, cross-fitted first stage. The L2-penalized logistic regression ($C = 1$) does not fit school dummies exactly, but shrinks them, preventing the nuisance model from mechanically absorbing all treatment variation in single-teacher-gender schools. The regularization is therefore necessary to stabilize the nuisance-function estimation and prevent separation, not to create identification ex nihilo. The causal interpretation still depends entirely on the selection-on-observables assumption holding across schools. This precision gain relative to OLS comes at the cost of that stronger requirement.

Result

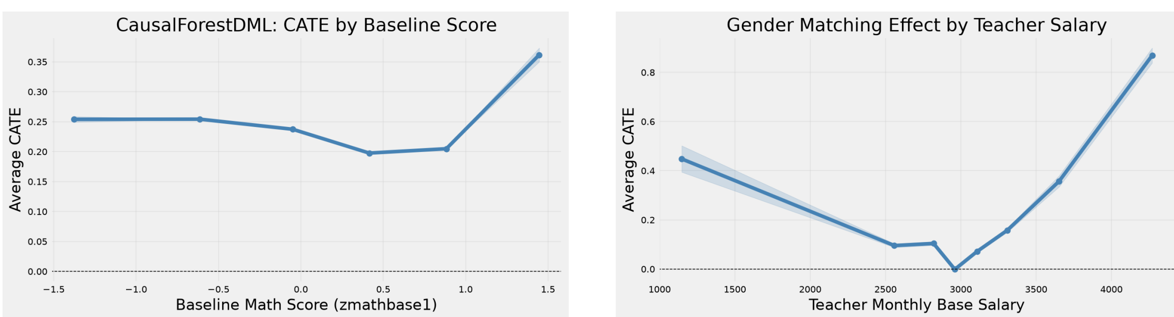
Estimated on the full sample, all three DML variants return significant positive average effects, and they agree closely (Table 8): the ATE sits between +0.213 and +0.230 SD, with confidence intervals that exclude zero. The direction matches the point estimates from the earlier sections; the magnitude is larger than the conservative +0.074–+0.098 SD range. I read this larger magnitude as evidence that the positive signal is not an artifact of the linear model, but also as the most model-dependent estimate in the paper because it relies on the richest control set and flexible nuisance-function estimation. Where the three specifications part ways is heterogeneity, not level. LinearDML and SparseLinearDML impose a functional form on the CATE — linear and cubic in the baseline score, respectively — and neither finds significant modifier structure: LinearDML's slope on the baseline score and every polynomial term in SparseLinearDML are individually insignificant. Only CausalForestDML, which is nonparametric in the effect modifier, recovers a heterogeneity pattern the data actually support, so the figures below use it rather than the parametric variants. These averages are best read alongside the conservative estimates rather than as replacements for them — what they add is flexible residualization and, below, the shape of the heterogeneity.

Figure 8 connects that heterogeneity back to the earlier findings with two causal forests that differ only in their effect-modifier set. Panel (a) uses a forest whose sole modifier is the baseline math score and bins the predicted CATEs by that score: the effect is roughly flat at +0.20–+0.25 SD across most of the distribution, rising only in the top bin. Baseline achievement therefore does not appear

Specification	**ATE (SD)**	**95% CI**
LinearDML (baseline score, linear)	+0.222	(0.066, 0.377)
SparseLinearDML (baseline score, cubic)	+0.213	(0.058, 0.369)
CausalForestDML (baseline score, nonparametric)	+0.230	(0.214, 0.247)

Table 8: Full-sample DML average treatment effects for female gender matching. All three specifications place the ATE between +0.21 and +0.23 SD with intervals excluding zero. The CausalForestDML interval is the doubly robust ATE interval; the other two are from the coefficient summary. These estimates are larger than the conservative range in earlier sections and should be interpreted under the stronger DML assumptions. Widening the modifier set to include teacher salary (CausalForestDML, 2D) gives +0.264 SD (0.244, 0.283).

to explain much of the treatment-effect variation. Panel (b) adds teacher base salary to the modifier set and bins the predictions by salary. A pronounced V shape appears, with CATEs around +0.45 SD in the lowest-salary classrooms, near zero around 3,000 yuan, and up to +0.87 SD at the top. This suggests that the predicted benefit of matching is largest for female students in the lowest- and highest-salary classrooms and close to zero for the modal middle. Because each individual CATE here depends jointly on salary and baseline score, this salary curve is a *marginal* average rather than a strict salary effect holding baseline fixed. Teacher salary and student baseline are essentially uncorrelated in these data ($r = -0.001$), and the effect is nearly flat in baseline (panel a), so the salary pattern is less likely to be only a re-sorting on baseline achievement. Figure 9 then lets every pre-treatment modifier compete in a single CATE tree: salary again takes the root split, ahead of the baseline scores that dominated the predictive trees in Section 3.3. This suggests that the variables predicting the level of achievement are not necessarily the same variables predicting the benefit of gender matching.



(a) CATE by baseline-score bin

(b) CATE by teacher monthly base salary

Figure 8: CausalForestDML heterogeneity. Panel (a): a causal forest with the baseline math score as its only effect modifier (roughly flat). Panel (b): a causal forest that adds teacher base salary to the modifier set, binned by salary (V-shaped). The parametric LinearDML and SparseLinearDML impose a functional form on the CATE and find no significant modifier structure, so they are omitted.

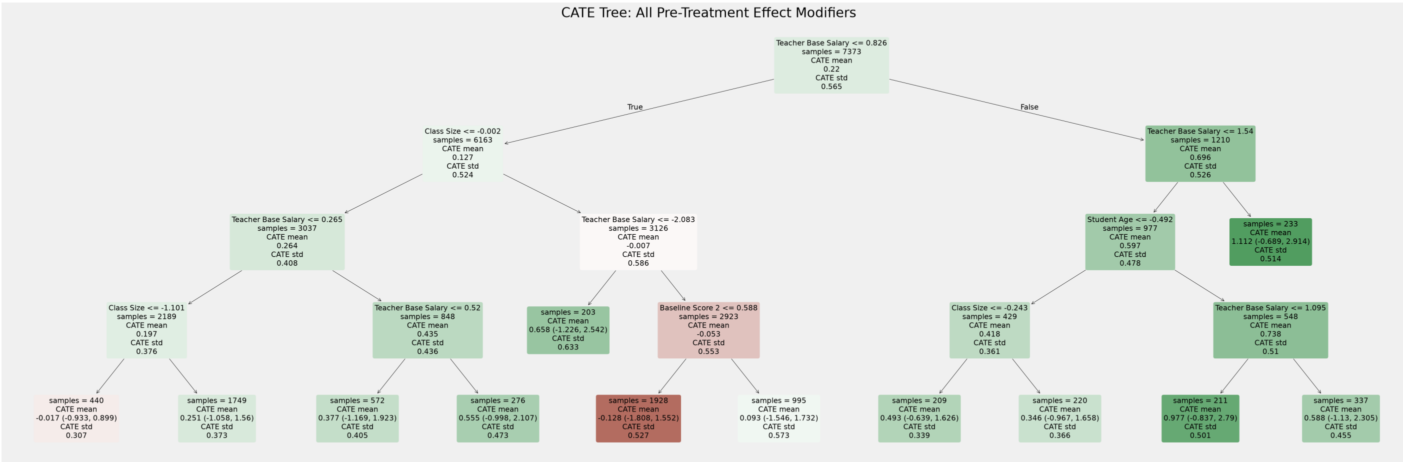


Figure 9: CATE decision tree over all pre-treatment effect modifiers (standardized units; root split on teacher base salary)

Beyond locating the modifiers, the full-modifier forest also describes how much the predicted effect varies across students. Figure 10 plots the distribution of its individual CATE estimates: they average +0.220 SD, but spread widely, with a standard deviation of 0.565 SD. Most students are predicted to benefit, while a thin left tail sits near or just below zero. This suggests that the matching benefit is not uniform across classrooms. One caveat is important: once nine modifiers compete for splits, the forest's own average is imprecisely estimated, so this distribution is read for shape rather than as a sharper estimate of the average effect.

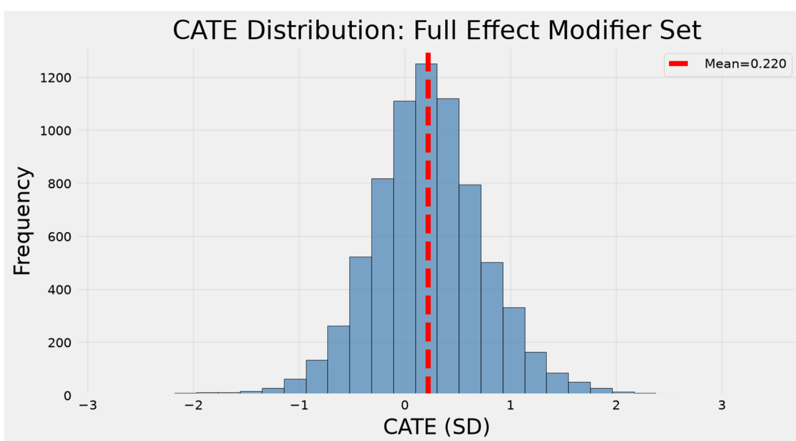


Figure 10: Distribution of individual CATE estimates from the causal forest with the full pre-treatment modifier set (mean +0.220 SD, standard deviation 0.565 SD).

4 Discussion of Results

The main finding is cautious but consistent: the female-female teacher-student match coefficient is positive across a range of specifications, while the size of the estimate depends on the baseline comparison and the identification assumptions. Under the most conservative within-school specification, the estimated coefficient is +0.077 SD. This corresponds to roughly 37% of the baseline gender gap, but it is imprecisely estimated because only 12 mixed-teacher schools identify the cell contrast. Broader and more model-dependent methods point in the same direction, but they should not be read as identical replications of the OLS coefficient.

The disciplined range from the four conservative methods is +0.074 to +0.098 SD. OLS uses only within-school variation and is therefore the most design-transparent but least precise. IPW reweights the full sample and gives +0.098 SD. The doubly robust estimator combines the same propensity score with a linear outcome model and gives +0.087 SD. The DAG backdoor estimator uses the graph-defined minimal adjustment set and gives +0.074 SD. These estimates differ in baseline and adjustment set, so the important point is not that they are numerically identical. The important point is that reasonable contrasts around the same substantive question all point in the positive direction. This is also how I read the meta-learners: the tuned T-learner is inside the conservative range, the X-learner is smaller but positive, and the S-learner is close to zero in a way that is consistent with the known weak-treatment problem of pooled S-learners. Because the meta-learners do not carry valid confidence intervals, they are useful as a directional check rather than as sharper estimates.

A mechanical feature of the data explains why the estimates differ in precision. The treatment is a product of two genders, `f_same_sex` = `female` × `t_female`, and teacher gender is very nearly a school-level constant: 202 of 214 schools have a single-gender teaching roster. Any method that absorbs school-level variation with unpenalized dummies therefore leaves only the 12 mixed-teacher schools to identify the cell contrast. This is exactly what happens in the fixed-effects OLS. The broader methods relax that restriction in different ways. The DAG backdoor does not condition on schools under the assumed graph; IPW and the doubly robust estimator use a reweighted full-sample comparison; and DML keeps the richest controls but uses regularized, cross-fitted nuisance models to prevent separation. The balance check of Section 2.3, pre-treatment controls, and county fixed effects make the broader comparisons more plausible, but they do not prove conditional independence across schools. For that reason, the +0.074 to +0.098 SD range remains the disciplined anchor, while DML is read as flexible upper-range evidence and as the main tool for heterogeneity analysis.

The DML estimates are larger than the conservative range: LinearDML gives +0.222 SD, SparseLinearDML gives +0.213 SD, and CausalForestDML gives +0.230 SD, all with confidence intervals excluding zero. One interpretation is that flexible nuisance-function estimation captures nonlinearities and interactions that the linear model does not. At the same time, DML also relies on the richest control set and on selection-on-observables across schools, so I do not treat these estimates as replacing the conservative range. They are most useful for showing that the positive signal is not limited to the linear specification and for studying where the estimated effect is larger or smaller.

The heterogeneity evidence points most strongly to teacher base salary, but I interpret that variable as a proxy rather than a policy lever. Binning CausalForestDML predictions by salary gives a non-monotonic pattern: predicted matching benefits are large in very low-salary classrooms and very high-salary classrooms, and close to zero in the middle of the distribution. This pattern is suggestive rather than definitive. In low-salary classrooms, a female math teacher may be especially salient as a role model if female representation is scarce. In high-salary classrooms, teacher-student interaction quality or lower gender-biased assessment may matter more. The data do not directly test these mechanisms, so I treat them as possible interpretations of the CATE pattern rather than as established channels.

The policy implication is therefore limited. The results suggest that female-female teacher-student matching may matter, and that the gains may be larger in particular classroom contexts. They do not show that raising salary would causally increase the matching effect, nor do they show that assigning every female student to a female math teacher would produce the same average gain. A more cautious implication is that teacher assignment and classroom context deserve further study together, especially in schools where female students have little exposure to female math teachers. Whether these effects persist into secondary school and affect later STEM choices remains outside the scope of this data.

5 Conclusion and Future Directions

This paper uses data from a teacher performance-pay RCT conducted in rural western China (Loyalka et al. 2019) to study teacher-student gender matching and female students' math achievement. The original experiment was not designed to randomize teacher gender, so the analysis relies on observational variation within the experimental sample. The most conservative fixed-effects estimate is positive but imprecise: the female-female matching coefficient is +0.077 SD, which corresponds to roughly 37% of the baseline gender gap, but the estimate is identified from only 12 mixed-teacher schools. Broader methods give similar positive estimates: IPW gives +0.098 SD, the doubly robust estimator gives +0.087 SD, and the DAG backdoor gives +0.074 SD. I interpret this +0.074 to +0.098 SD range as the disciplined range of the paper.

Double Machine Learning gives larger estimates, from +0.213 to +0.230 SD, under the richest control set and flexible residualization. I treat these estimates as upper-range evidence under stronger assumptions and as the basis for heterogeneity analysis. The CausalForestDML results suggest that teacher base salary is the main observed modifier, with larger predicted benefits in very low- and very high-salary classrooms and smaller predicted benefits in the middle. Because salary is not randomized, this pattern should be interpreted as evidence about classroom conditions correlated with salary, not as proof that salary itself causally moderates the effect.

The main contribution of the paper is therefore not a single definitive estimate. It is a structured comparison of related estimates around the same substantive question: whether female-female math teacher-student matching is associated with better outcomes for female students. Across conservative linear models, graph-based adjustment, propensity-score methods, and flexible machine learning, the sign is consistently positive. The main limitations are the short follow-up window and the lack of randomized teacher assignment. Future work with randomized or quasi-random teacher assignment, longer follow-up data, and direct measures of classroom mechanisms would help determine whether these estimated achievement gains persist and whether they translate into later STEM course-taking or occupational choices.

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7 Appendix

Table 9: Balance Check, Complete-Case Sample (N = 7,744)

	Male teacher	Female teacher	Difference	SMD	Within-school coef.	p
Math baseline 1	−0.140	0.134	0.274	0.271	0.0026	0.100
Math baseline 2	−0.084	0.146	0.229	0.223	0.0013	0.398
Student age	12.056	11.968	−0.088	−0.086	0.0013	0.408
SES index	−0.644	−0.547	0.097	0.095	−0.0021	0.168
Father's education (JHS+)	0.503	0.567	0.064	0.129	0.0004	0.899
Mother's education (JHS+)	0.320	0.363	0.043	0.091	−0.0000	0.991

Note: All columns are computed on the same complete-case sample of 7,744 students for whom all six baseline characteristics are non-missing, so the means here differ slightly from those in Table 2, which computes each mean on all observations available for that characteristic. SMD is the standardized mean difference (raw difference divided by the pooled standard deviation). The within-school coefficient is the estimate on each characteristic in a separate OLS regression of teacher gender on that characteristic with school fixed effects and HC1 robust standard errors; it is a within-school slope, not a mean difference, and is close to zero for every characteristic. This is the source of the near-zero within-school association reported in Table 2: the sizeable raw differences (e.g. 0.274 SD in baseline math) reflect between-school sorting and all but vanish once school fixed effects are absorbed.

Figure 11: Original LASSO for Gender-Matching Model

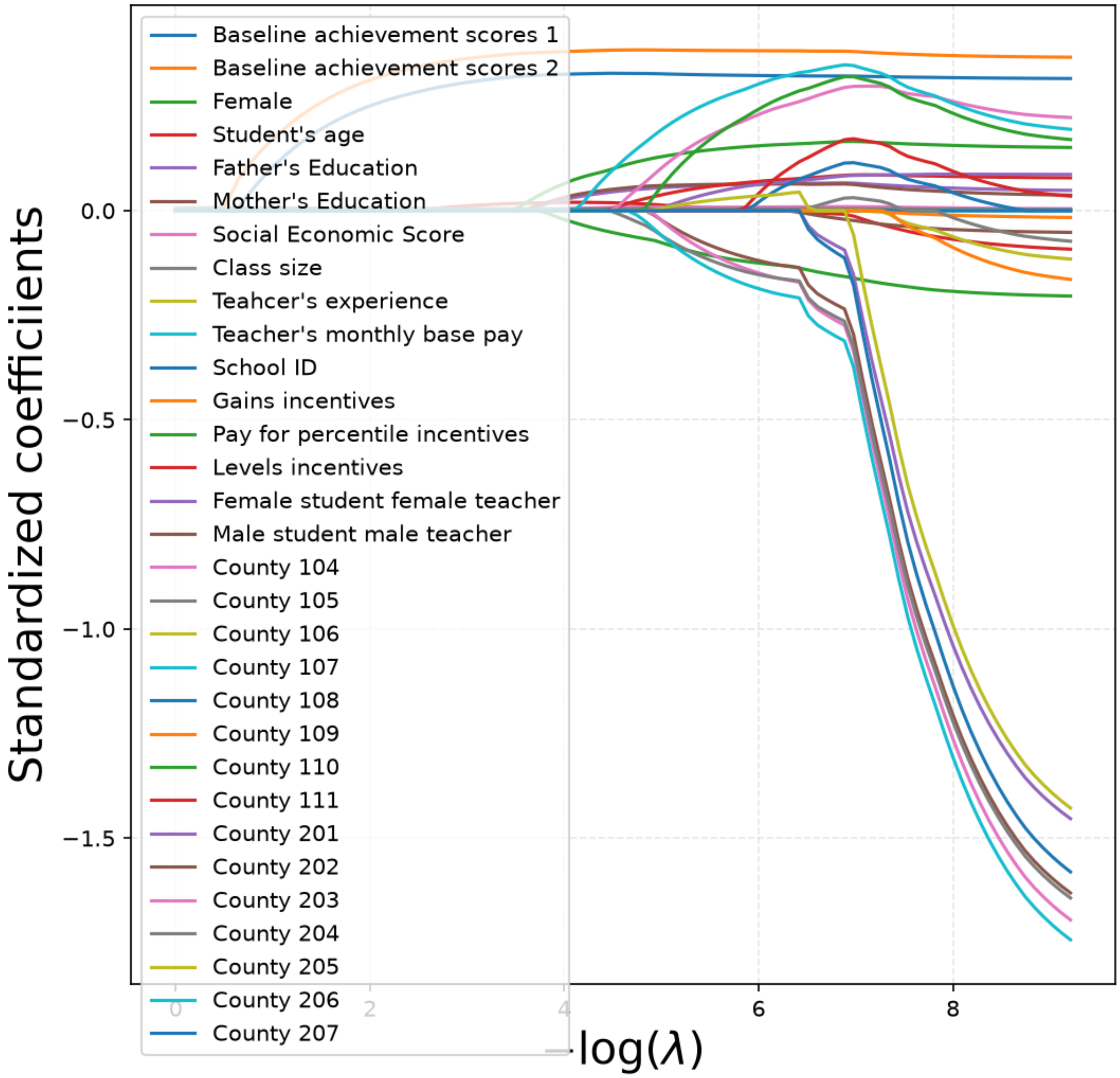


Figure 12: Original LASSO for Base Model

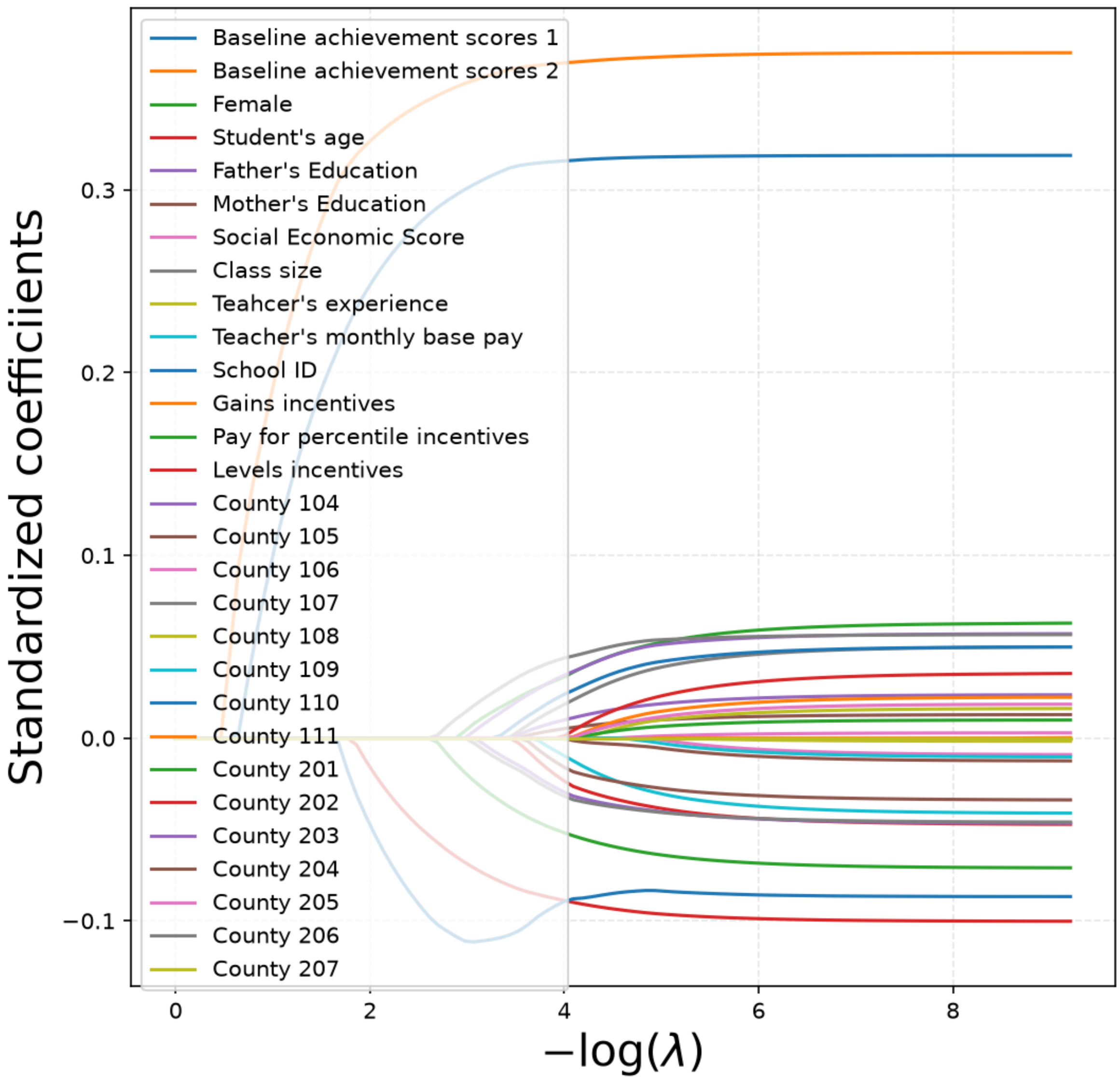


Figure 13: Original Ridge for Base Model

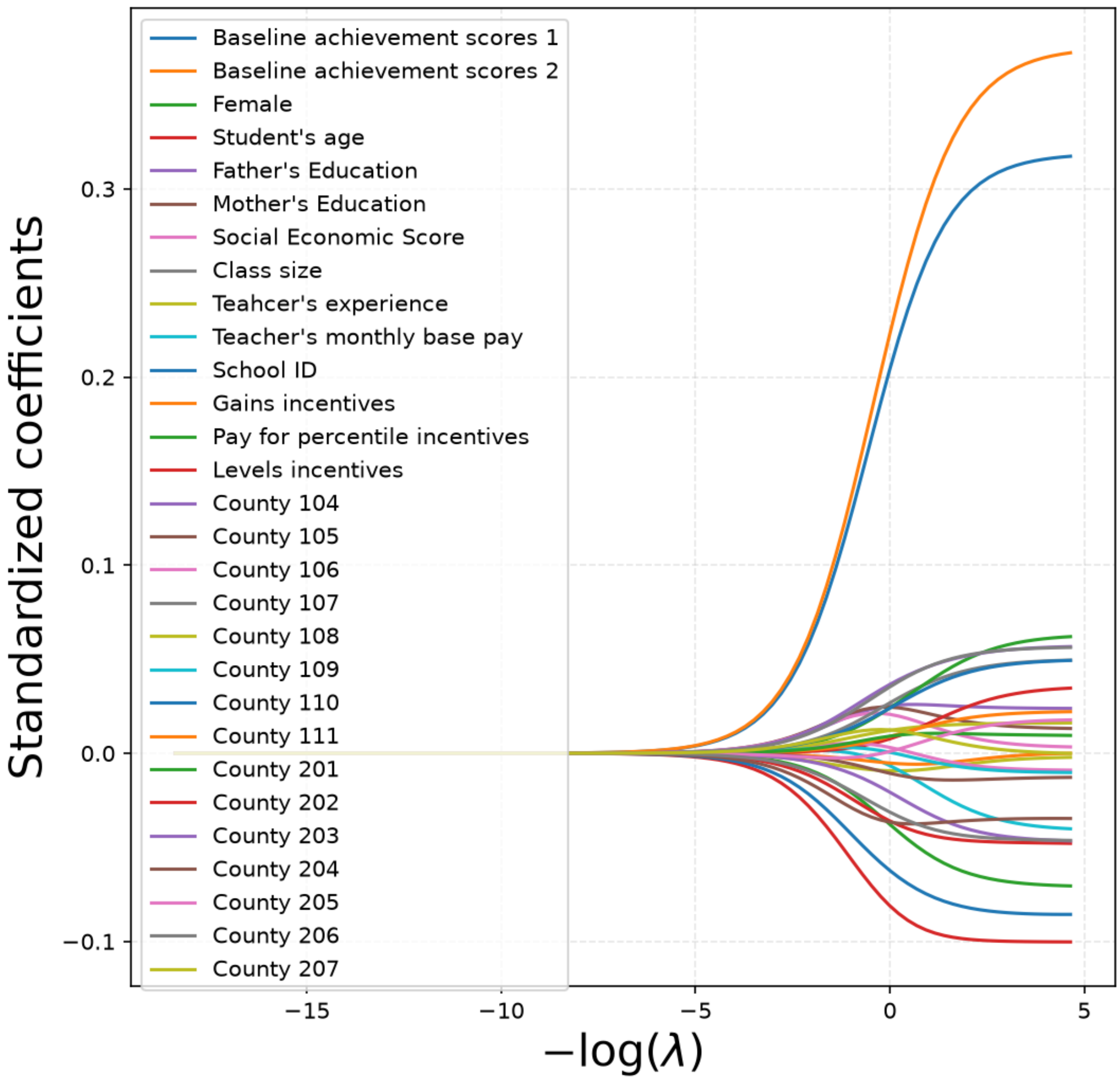
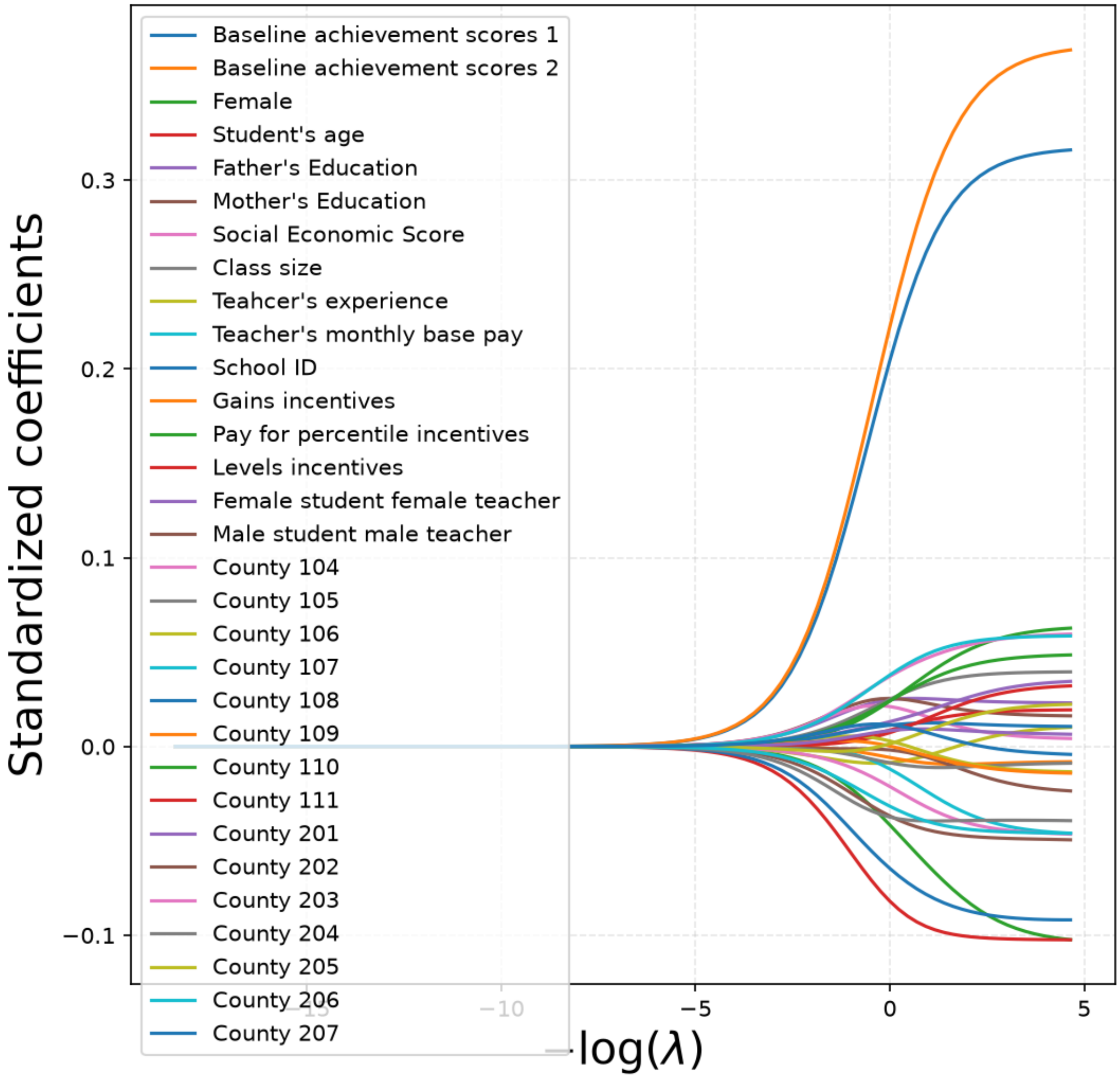


Figure 14: Original Ridge for Gender-Matching Model



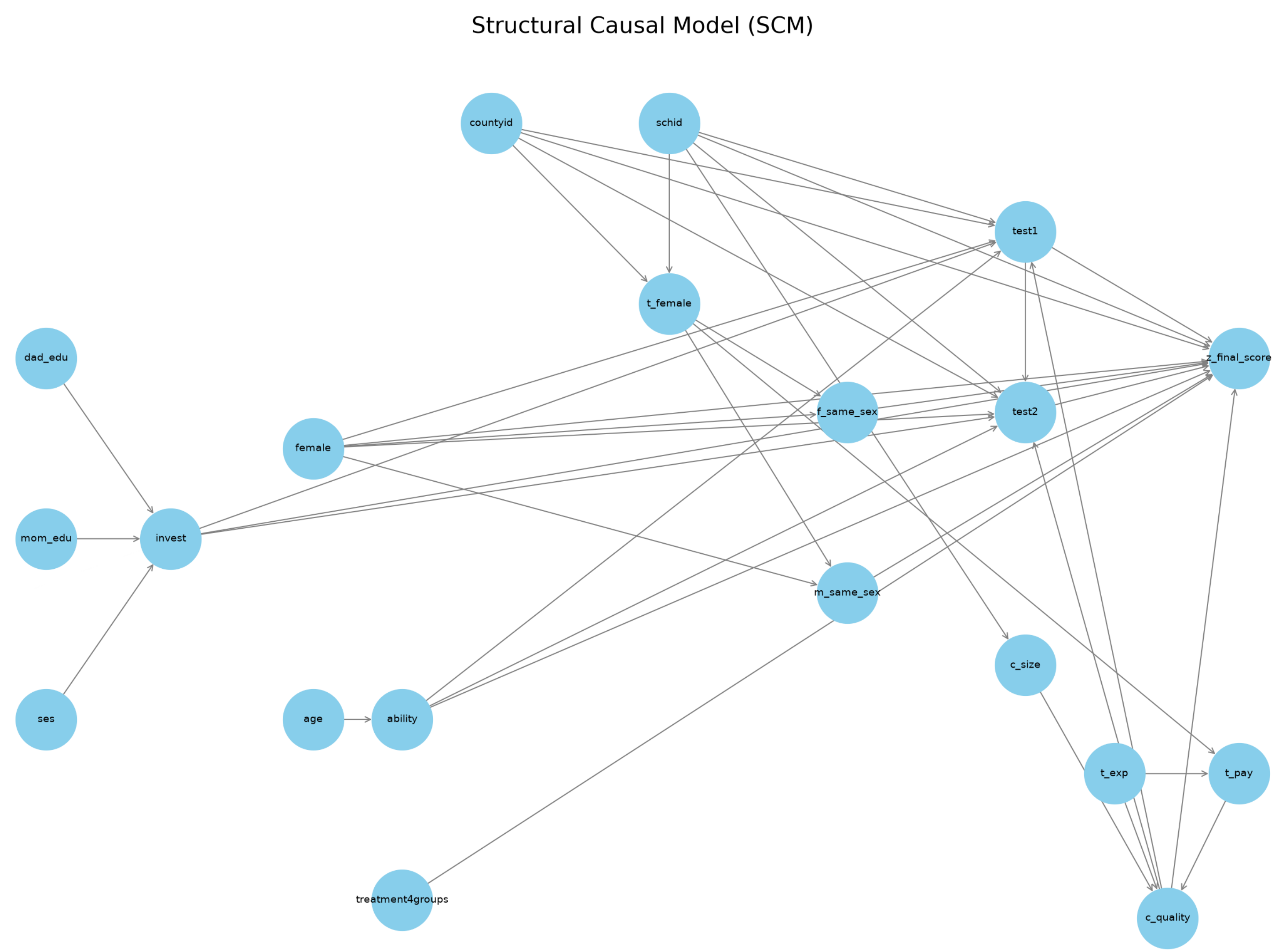


Figure 15: Original DAG: full structural causal model